

Q.1 →

Selenium webdriver is an open source automation framework used to automate web browser interactions, it directly communicates with browser and simulates real user actions like:

- clicking buttons
- entering text
- selecting dropdown options
- navigating between pages
- validating UI elements

It is mainly used for automated web application testing (functional, regression, cross browser testing).

Q.2 →

normally we can:-

- (i) download browser drivers manually.
- (ii) match driver version with browser version.
- (iii) configure system path

webdriver manager:-

- (i) automatically download the correct driver.
- (ii) manages version compatibility.
- (iii) Removes manual configuration.

Q.3 →

e.g. login test

Steps

- ① open website
- ② enter username & password
- ③ click login
- ④ Verify dashboard page loads.

best practice:

- Use page object model (pom)
- separate test logic & locators
- use proper waits.

Q.4 →

best locators

- ID
- Name
- CSS selector
- XPath → For dynamic elements

For dynamic elements:

- Use contains () or starts-with () in XPath
- prefer explicit wait

best wait

- i] Explicit wait [recommended]
- ii] Implicit wait [basic use]
- iii] Fluent wait [advance use]

Q.3 →

Use assertions to compare expected vs actual results.

e.g.

- Verify page title
- Verify url
- Verify element is displayed.
- Verify success message text.

e.g. (java)

```
Assert.assertEquals(driver.getTitle(), "Dashboard");
```

IF assertion pass → test successful

IF assertion fails → test fails.